

1. Create a new ADLS Gen2 storage acc name - muharafcapstoneprj1

Home > Storage accounts > Create a storage account

Basics Advanced Networking Data protection Encryption Tags **Review + create**

[View automation template](#)

Basics

Subscription	LabsKraft
Resource group	LabsKraft2027
Location	Qatar Central
Storage account name	muharafcapstoneprj1
Primary service	Azure Blob Storage or Azure Data Lake Storage Gen 2
Performance	Standard
Replication	Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

Previous Next **Create** [Give feedback](#)

2. In the storage account go to Data Sharing > Container, and add a new container named *raw*

Home > muharafcapstoneprj1_1756096954205 | Overview > muharafcapstoneprj1

muharafcapstoneprj1 | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	
☐	\$logs	8/25/2025, 10:15:05 AM	Private	Available	...
☐	raw	8/25/2025, 10:16:27 AM	Private	Available	...

Navigation: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Partner solutions, Resource visualizer, Data storage, Containers, File shares, Queues, Tables, Security + networking, Networking

3. Upload the downloaded `retails_order.csv` file in the newly created raw container

The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation indicates the path: Home > muharafcapstoneprj1_1756096954205 | Overview > muharafcapstoneprj1 | Containers > raw. The 'raw' container is selected, and the 'Overview' tab is active. The 'Upload' button is highlighted in the top toolbar. Below the toolbar, the authentication method is 'Access key'. A search bar for blobs is present. A table shows 'Showing all 1 items' with the following data:

Name	Last modified	Access tier	Blob type	Size	Lease state
retail_iot_data.csv	8/25/2025, 10:17:59 AM	Hot (Inferred)	Block blob	130.76 KiB	Available

4. No go to the Access Connector for Azure Databricks, and create a new connect

The screenshot shows the 'Create an Access Connector for Azure Databricks' wizard in the Microsoft Azure portal. The 'Review + create' tab is selected. The wizard displays the following configuration details:

Basics	
Subscription	LabsKraft
Resource group	LabsKraft2027
Name	muharaf-connector-capstone-prj-1
Region	Qatar Central

Managed Identity	
Identity type	SystemAssigned

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

🔔

⚙️

?

🗨️

tredence11@michaelklo...
DEFAULT DIRECTORY (MICHAELK...

Home > AccessConnectorCreate-20250825102048 | Overview >

muharaf-connector-capstone-prj-1 ⚙️ ☆ ...
Access Connector for Azure Databricks

Search

Refresh

Delete

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Automation

Help

Essentials

Resource group (move)
[LabsKraft2027](#)

Location
Qatar Central

Subscription (move)
[LabsKraft](#)

Subscription ID
d008c7cc-9795-4292-bf79-74ae52cda63d

Tags (edit)
[Add tags](#)

State
Succeeded

Resource ID
/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/...

JSON View

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

🔔

⚙️

?

🗨️

tredence11@michaelklo...
DEFAULT DIRECTORY (MICHAELK...

Home >

Access Connector for Azure Databricks ... Filter Access Connectors by resource group and location. How many Access Connectors do I have in total? +1

Default Directory (michaelkoudkraft@hotmail.onmicrosoft.com)

+ Create ⚙️ Manage view ▾ 🔄 Refresh ⬇️ Export to CSV 🔗 Open query | 🔗 Assign tags 🗑️ Delete

Group by none ▾

Create new access connector for azure databricks [experience. Click here to access the old experience.](#)

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

+ Add filter

☐	Name ↑		Resource Group	Subscription	Subscription ID	Location	State
☐	basab_connect_adb	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	East US	Succeeded
☐	capstone_retail-connector-sdatta	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	Australia Central	Succeeded
☐	hetul_connector_01	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	Central India	Succeeded
☐	sagarconnector	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	West US 2	Succeeded
☐	sanyam-connector	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	West US 2	Succeeded
☐	shivam-connector	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	Japan East	Succeeded
☐	vaishnavi_connect	...	LabsKraft2027	LabsKraft	/subscriptions/d008c7c...	South India	Succeeded

Showing 1 - 7 of 7. Display count: auto ▾

Give feedback

https://portal.azure.com/#blade/Microsoft_Azure_Databricks/AccessConnectorCreate_dx

5. Now in the storage account, give the Access Control IAM

Home > muharafcapstoneprj1 | Access Control (IAM) >

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Search: blob Type: All Category: All

Name	Description	Type	Category	Details
CreateBlobStorage		CustomRole	None	View
Defender CSPM Storage Data Scan...	Grants access to read blobs and files. This role is used by the data scanner of Dfender CSPM.	BuiltInRole	None	View
Defender for Storage Data Scanner	Grants access to read blobs and update index tags. This role is used by the data scanner of...	BuiltInRole	None	View
Storage Blob Contributor	Allows for read, write and delete access to Azure Storage blob containers and data	BuiltInRole	Storage	View
Storage Blob Data Owner	Allows for full access to Azure Storage blob containers and data, including assigning POSI...	BuiltInRole	Storage	View
Storage Blob Data Reader	Allows for read access to Azure Storage blob containers and data	BuiltInRole	Storage	View
Storage Blob Delegator	Allows for generation of a user delegation key which can be used to sign SAS tokens	BuiltInRole	Storage	View

Showing 1 - 7 of 7 results.

[Review + assign](#) [Previous](#) [Next](#) [Feedback](#)

Home > AccessConnectorCreate-20250825102048 | Overview >

muharaf-connector-capstone-prj-1

Access Connector for Azure Databricks

Search Refresh Delete

Overview Essentials JSON View

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Automation

Help

Resource group (move) [LabsKraft2027](#)

Location: Qatar Central

Subscription (move) [LabsKraft](#)

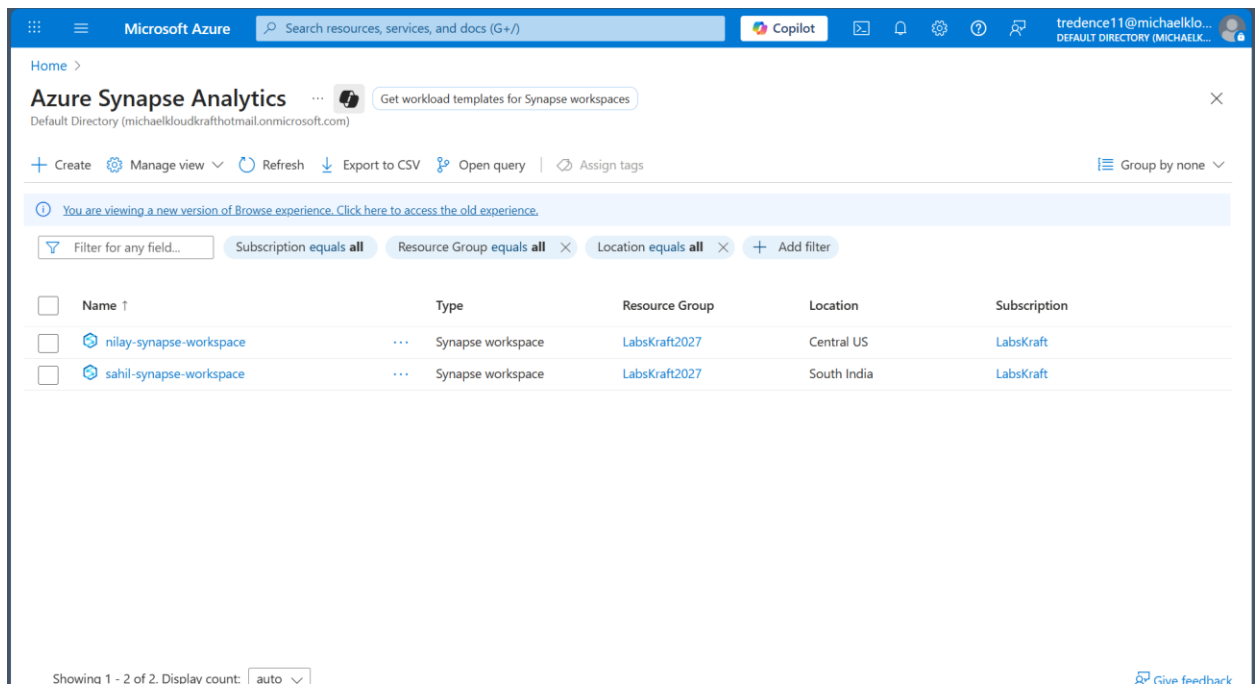
Subscription ID: d008c7cc-9795-4292-bf79-74ae52cda63d

Tags (edit) [Add tags](#)

State: Succeeded

Resource ID: /subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/...

6. Now create a Azure Synapse Workspace and then open the Synapse Studio

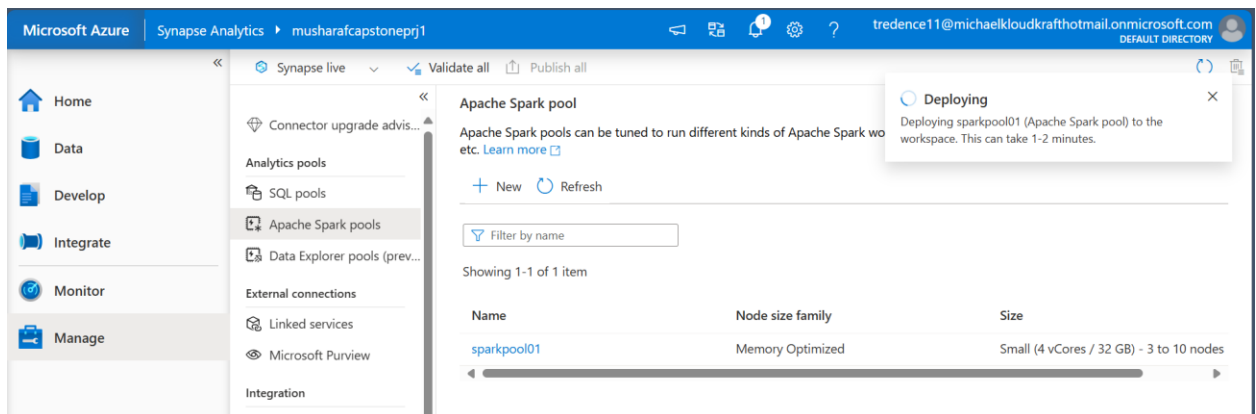


The screenshot shows the Microsoft Azure portal interface for Azure Synapse Analytics. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile. The main content area displays the 'Azure Synapse Analytics' page with a 'Create' button and a 'Manage view' dropdown. A table lists the existing Synapse workspaces:

Name	Type	Resource Group	Location	Subscription
nilay-synapse-workspace	Synapse workspace	LabsKraft2027	Central US	LabsKraft
sahil-synapse-workspace	Synapse workspace	LabsKraft2027	South India	LabsKraft

At the bottom, it indicates 'Showing 1 - 2 of 2. Display count: auto'.

7. At this stage, create a new Apache Spark Pool



The screenshot shows the Microsoft Azure Synapse Studio interface. The left sidebar contains navigation links: Home, Data, Develop, Integrate, Monitor, and Manage. The main content area displays the 'Apache Spark pool' configuration page. A 'Deploying' notification is visible, stating: 'Deploying sparkpool01 (Apache Spark pool) to the workspace. This can take 1-2 minutes.' The table below shows the configuration for the 'sparkpool01' pool:

Name	Node size family	Size
sparkpool01	Memory Optimized	Small (4 vCores / 32 GB) - 3 to 10 nodes

8. Now Search for Azure databricks > tred_d3 > Launch Workspace

Microsoft Azure Search resources, services, and docs (G+)

Home > Azure Databricks >

tred_d3 Azure Databricks Service

Show me the cluster details for this Azure Databricks Service. List access control settings for this Azure Databricks Service. +1

Search Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Active

Resource group: [LabsKraft2027](#)

Location: Central US

Subscription: [LabsKraft](#)

Subscription ID: d008c7cc-9795-4292-bf79-74ae52cda63d

Tags ([edit](#)) [Add tags](#)

[databricks-rg-tred_d3-bkeujwrg6ffdg](#)

URL: <https://adb-1405430580087508.8.azuredatabricks.net>

Pricing Tier: [Premium \(+ Role-based access controls\). \(Click to change\)](#)

Enable No Public IP: Yes



[Launch Workspace](#)

[Documentation](#) [Getting Started](#) [Import Data from File](#) [Import Data from Azure Storage](#)

9. In the Databricks > Catalogue > Create Catalogue

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

tred_d3

Create a new catalog

A catalog is the first layer of Unity Catalog's three-level namespace and is used to organize your data assets. [Learn more](#)

Catalog name*:

Type*:

Storage location:

[Create a new external location](#)

Location in cloud storage where data for managed tables will be stored. If not specified, the location will default to the metastore root location.

[Cancel](#) [Create](#)

Filter

Last viewed: 3 days ago

Type: Catalog

[Send feedback](#)

Catalog

Serverless Starter Warehouse Serverless 5

Type to search...

- My organization
 - tred_d3
 - system
 - ishika_c
 - musharaf
 - ishika_c
 - ishika_r
 - nilay_c
 - nilay_n
- Delta Shares R
- Legacy
 - hive_m

Workspace

Recents

Catalog

Jobs & Pipelines *

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

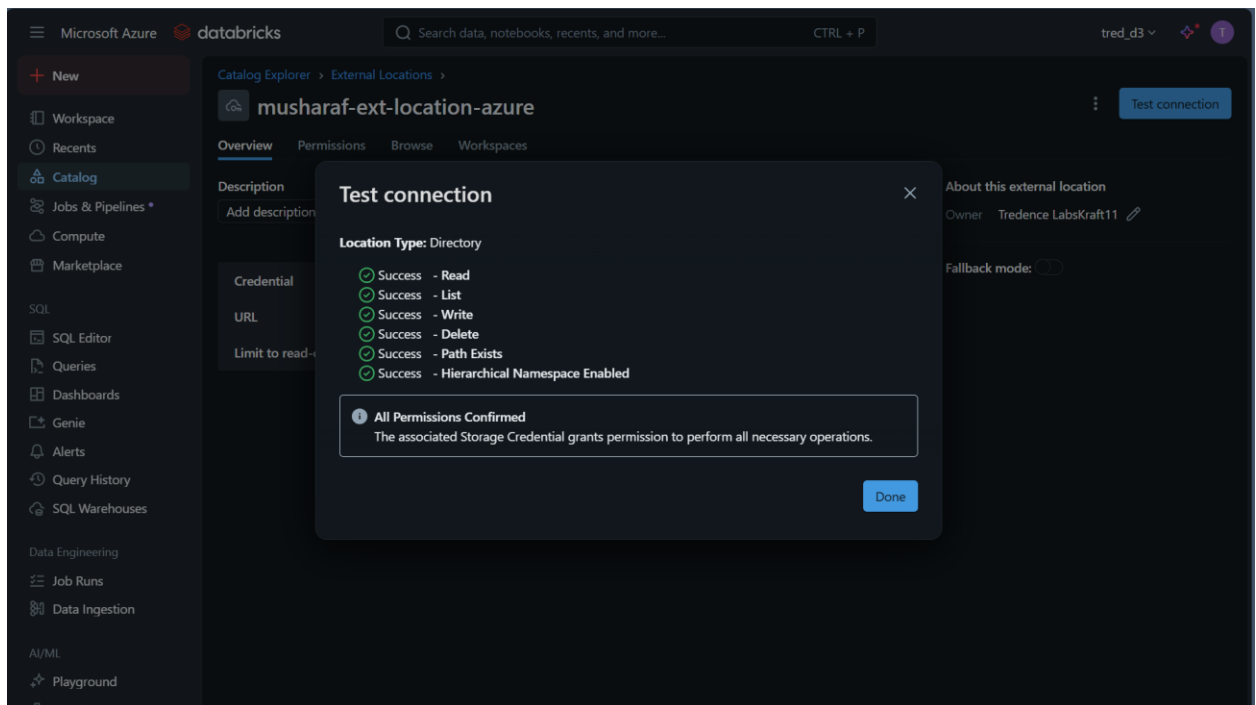
Data Ingestion

AI/ML

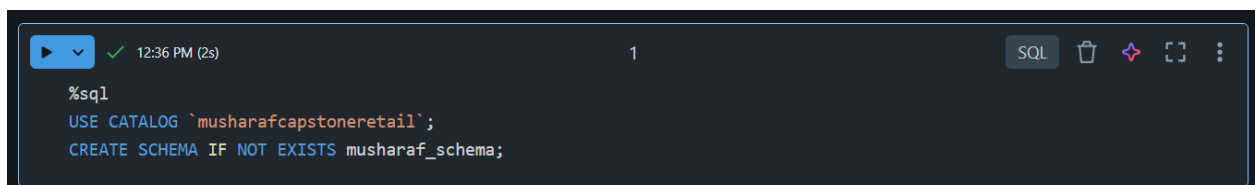
Playground

Experiments

10. Create a new external location, and then after creating test connection



11. After testing the connection, create a new notebook and add the following commands



12. After creating a new Schema, now create a new notebook and add the following code:

```

1 from dlt import table
2 import pyspark.sql.functions as F
3 from pyspark.sql.types import StructType, StructField, StringType, DoubleType, IntegerType, TimestampType
4
5 # :white_check_mark: Define schema for retail IoT CSV
6 iot_schema = StructType([
7     StructField("OrderID", StringType(), True),
8     StructField("CustomerID", StringType(), True),
9     StructField("Region", StringType(), True),
10    StructField("StoreID", StringType(), True),
11    StructField("Product", StringType(), True),
12    StructField("Quantity", IntegerType(), True),
13    StructField("UnitPrice", DoubleType(), True),
14    StructField("SalesAmount", DoubleType(), True),
15    StructField("OrderTime", StringType(), True), # can cast to Timestamp later in Silver
16    StructField("DeviceType", StringType(), True),
17    StructField("Temperature", DoubleType(), True),
18    StructField("DeviceStatus", StringType(), True),
19    StructField("AnomalyFlag", IntegerType(), True),
20    StructField("AnomalyType", StringType(), True)
21 ])
22 # :white_check_mark: Paths (use subdirectories to avoid overlap with DLT system files)
23 raw_path = "abfss://raw@muharafcapstoneprj1.dfs.core.windows.net/raw_data/"
24 schema_path = "abfss://raw@muharafcapstoneprj1.dfs.core.windows.net/dlt_schemas/bronze_orders_musharaf"
25 checkpoint_path = "abfss://raw@muharafcapstoneprj1.dfs.core.windows.net/dlt_checkpoints/bronze_orders_musharaf"
26 @table(
27     name="musharafcapstoneretail.musharaf_schema.bronze_orders",
28     comment="Bronze table - raw ingested IoT retail data via Auto Loader"
29 )
30 def bronze_orders():
31     return (
32         spark.readStream
33             .format("cloudFiles")
34             .option("cloudFiles.format", "csv")
35             .option("header", True)
36             .option("cloudFiles.schemaLocation", schema_path)
37             .schema(iot_schema) # :white_check_mark: manually set schema (no infer needed)
38             .load(raw_path) # :white_check_mark: points to /raw_data/ folder
39     )

```

```

2
Python

from dlt import table
import pyspark.sql.functions as F
from pyspark.sql.types import DoubleType, IntegerType

# =====
# Silver Layer - Cleaned IoT Orders
# =====
@table(
    name="musharafcapstoneretail.musharaf_schema.silver_orders",
    comment="Silver layer: cleaned and deduplicated IoT orders with anomaly handling"
)
def silver_orders():
    # Read Bronze
    orders = dlt.read("bronze_orders")
    # Standardize schema
    orders = (
        orders.withColumn("Quantity", F.col("Quantity").cast(IntegerType()))
        .withColumn("UnitPrice", F.col("UnitPrice").cast(DoubleType()))
        .withColumn("SalesAmount", F.col("SalesAmount").cast(DoubleType()))
        .withColumn("Temperature", F.col("Temperature").cast(DoubleType()))
        .withColumn("OrderTime", F.to_timestamp("OrderTime", "M/d/yyyy H:mm"))
    )
    # Deduplicate by OrderID
    orders = orders.dropDuplicates(["OrderID"])
    # Handle missing anomaly type
    orders = orders.withColumn(
        "AnomalyType",
        F.when((F.col("AnomalyFlag") == 1) & (F.col("AnomalyType").isNull()), "Unknown")
        .when((F.col("AnomalyFlag") == 1) & (F.col("AnomalyType") == ""), "Unknown")
        .otherwise(F.col("AnomalyType"))
    )
    # Add business rule fields
    orders = (
        orders.withColumn("IsAnomaly", F.col("AnomalyFlag") == 1)
        .withColumn("DeviceStatus", F.upper(F.col("DeviceStatus")))
        .withColumn("NormalizedStoreID", F.upper(F.col("StoreID")))
        .withColumn("NormalizedProduct", F.upper(F.col("Product")))
    )
    return orders

```



```

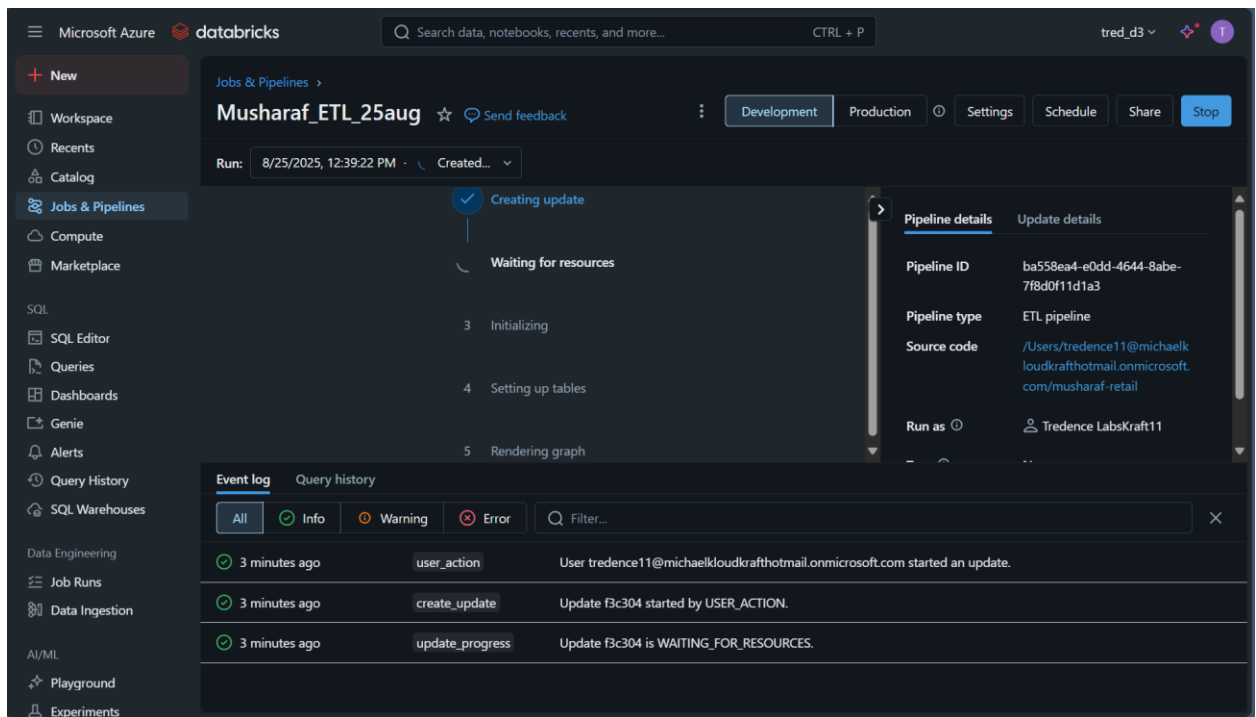
1 from dlt import table
2 import pyspark.sql.functions as F
3 from pyspark.sql.types import StructType, StructField, StringType, DoubleType, IntegerType, TimestampType
4
5 # :white_check_mark: Define schema for retail IoT CSV
6 iot_schema = StructType([
7     StructField("OrderID", StringType(), True),
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9     StructField("Region", StringType(), True),
10    StructField("StoreID", StringType(), True),
11    StructField("Product", StringType(), True),
12    StructField("Quantity", IntegerType(), True),
13    StructField("UnitPrice", DoubleType(), True),
14    StructField("SalesAmount", DoubleType(), True),
15    StructField("OrderTime", StringType(), True), # can cast to Timestamp later in Silver
16    StructField("DeviceType", StringType(), True),
17    StructField("Temperature", DoubleType(), True),
18    StructField("DeviceStatus", StringType(), True),
19    StructField("AnomalyFlag", IntegerType(), True),
20    StructField("AnomalyType", StringType(), True)
21 ])
22 # :white_check_mark: Paths (use subdirectories to avoid overlap with DLT system files)
23 raw_path = "abfss://raw@muharafcapstoneprj1.dfs.core.windows.net/raw_data/"
24 schema_path = "abfss://raw@muharafcapstoneprj1.dfs.core.windows.net/dlt_schemas/bronze_orders_musharaf"
25 checkpoint_path = "abfss://raw@muharafcapstoneprj1.dfs.core.windows.net/dlt_checkpoints/bronze_orders_musharaf"
26 @table(
27     name="musharafcapstoneretail.musharaf_schema.bronze_orders",
28     comment="Bronze table - raw ingested IoT retail data via Auto Loader"
29 )
30 def bronze_orders():
31     return (
32         spark.readStream
33             .format("cloudFiles")
34             .option("cloudFiles.format", "csv")
35             .option("header", True)
36             .option("cloudFiles.schemaLocation", schema_path)
37             .schema(iot_schema) # :white_check_mark: manually set schema (no infer needed)
38             .load(raw_path) # :white_check_mark: points to /raw_data/ folder
39     )

```

13. After adding the above code, go to the Jobs Pipelines, create a new Pipeline add the path as the same notebook you have added the above code into and then start the pipeline.

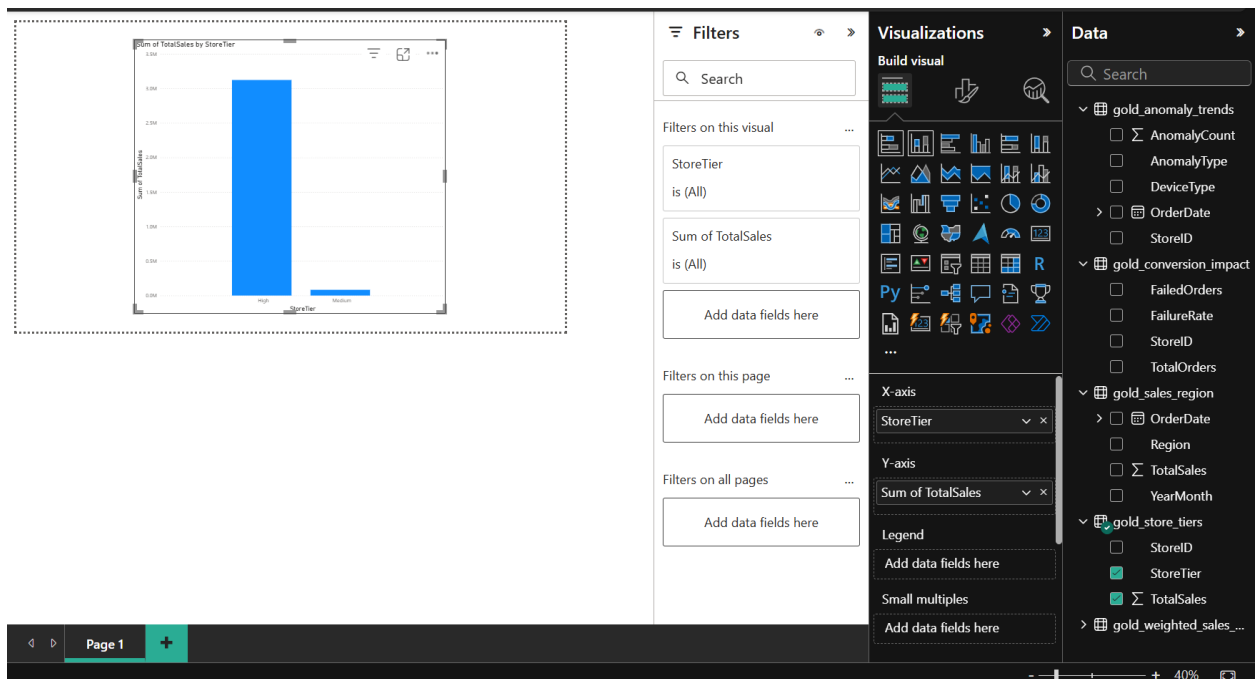
The screenshot displays the Databricks Jobs & Pipelines interface. The pipeline named 'Musharaf_ETL_25aug' is shown in the 'Development' environment. The pipeline is currently in the 'Waiting for resources' state. The interface includes a sidebar with navigation options like Workspace, Recents, Catalog, and Jobs & Pipelines. The main area shows the pipeline's progress through five steps: Creating update, Waiting for resources, Initializing, Setting up tables, and Rendering graph. The 'Event log' at the bottom shows recent events, including the pipeline being started and its current state.

Event	Time	Message
user_action	3 minutes ago	User tredence11@michaelkloudkraft@hotmail.onmicrosoft.com started an update.
create_update	3 minutes ago	Update f3c304 started by USER_ACTION.
update_progress	3 minutes ago	Update f3c304 is WAITING_FOR_RESOURCES.



14. At this stage, the medallion layers have been successfully created and now we will be using Power BI to create visualizations for the same gold layer data.

15. Go to the **SQL Warehouses**, Once running, click on the warehouse name. The **Open Power BI Desktop**. Click on **"Get Data" -> "More..."**, and search for or select **"Azure Databricks"** from the list. Click **"Connect"** and in the dashboard create a graph for the table you want to visualize



16. At this stage, create your VM

Microsoft Azure Search resources, services, and docs (G+/)

Home > **Create a virtual machine** ...

Help me create a VM optimized for high availability Help me choose the right VM size for my workload Help me create a low cost VM

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Price

1 X Standard B1s
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
1.2063 INR/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Preferred e-mail address

Preferred phone number

< Previous Next > **Create**

[Download a template for automation](#) [Give feedback](#)

17. Now create a new dedicated SQL Pool

Microsoft Azure Search resources, services, and docs (G+/)

Home > Azure Synapse Analytics > musharafcapstoneprj1 > **New dedicated SQL pool** ...

* Basics * Additional settings Tags Review + create

Create a dedicated SQL pool with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Dedicated SQL pool details

Name your dedicated SQL pool and choose its initial settings.

Dedicated SQL pool name *

Performance level ⓘ

Estimated price ⓘ **Est. Cost Per Hour**
1405.99 INR
[View pricing details](#)

Review + create Next: Additional settings >

18. Create Azure Event Hubs

The screenshot displays the Microsoft Azure portal interface. At the top, the header includes the Microsoft Azure logo, a search bar, and the user profile 'tredence11@michaelkio...'. The main content area shows the 'Overview' page for the Event Hubs Namespace 'musharafcapstoneproj1'. A left-hand navigation pane lists various Azure services. The main content area includes a search bar, action buttons (Event Hub, Delete, Refresh, Give feedback), and a notification banner about the Azure Event Hubs Data Explorer. Below this, the 'Essentials' section provides key details about the namespace, including its resource group, status, location, subscription, and host name. A table of properties is also visible, showing details like creation time, update time, zone redundancy, pricing tier, throughput units, and local authentication. At the bottom, there are summary cards for 'NAMESPACE CONTENTS' (0 Event Hubs), 'KAFKA SURFACE' (Not Supported), and 'ZONE REDUNDANCY' (Enabled). A 'Show data for the last:' filter is set to '1 hour'.

Microsoft Azure Search resources, services, and docs (G+/)

Home > musharafcapstoneproj1 Event Hubs Namespace

Search Event Hub Delete Refresh Give feedback

You can start generating test data or inspect data that has already been sent with the new Azure Event Hubs Data Explorer. Click on this message to try the feature!

Essentials JSON View

Resource group	(move) LabsKraft2027	Created	Monday, August 25, 2025 at 10:52:59 GMT+5:30
Status	Succeeded	Updated	Monday, August 25, 2025 at 10:53:40 GMT+5:30
Location	Qatar Central	Zone Redundancy	Enabled
Subscription	(move) LabsKraft	Pricing tier	Basic
Subscription ID	d008c7cc-9795-4292-bf79-74ae52cda63d	Throughput Units	1 unit
Host name	musharafcapstoneproj1.servicebus.windows.net	Auto-inflate throughput units	Not Supported
		Local Authentication	Enabled

Tags (edit) Add tags

NAMESPACE CONTENTS	KAFKA SURFACE	ZONE REDUNDANCY
0 EVENT HUBS	NOT SUPPORTED	ENABLED

Show data for the last: 1 hour 6 hours 12 hours 1 day 7 days 30 days

Add or remove favorites by pressing Ctrl+L+Shift+F